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Research Proposal — Track 1, Theme 1: Foundational ML (Model Innovation / RL Architectures)

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TITLE OF PROJECT: Safe, Sample-Efficient Robot Manipulation: Structured Representations, Reward Engineering, Temporal Intelligence, and Formal Safety Guarantees

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ABSTRACT

Making robots operate reliably in complex, unstructured environments demands progress on four interconnected fronts that current methods address only in isolation. **First**, the transition from safe offline data to effective online learning is unstable: exploratory policies deviate from the behavioural support of demonstrations and suffer catastrophic forgetting, while methods that constrain exploration to a learned latent manifold encounter a principled performance ceiling — the *exploitation gap* — that limits their asymptotic performance. **Second**, the design of dense reward functions is empirical folklore; the geometric properties of a reward function have a causal, first-order effect on policy gradient dynamics that is neither theoretically understood nor systematically studied. **Third**, current neural architectures treat time as a sequential counter rather than as a structured dimension of experience: they cannot reason about how long a subtask should take, anticipate when a strategy should change, or build plans that respect temporal constraints — a fundamental gap in machine intelligence that limits long-horizon task performance. **Fourth**, real-world deployment requires safety guarantees, yet the constraints governing robot behaviour are often hard to formalise: they may be latent cost functions, logical rules expressible in temporal logic, control-theoretic forbidden sets, or ill-defined human preferences — and no unified framework currently bridges these paradigms.

This proposal presents a four-pronged research programme that addresses each of these challenges in a principled and interconnected way. Building on our group's recent work on offline-to-online RL with structured latent action representations (SPAARS, under review at IROS 2026), we propose to extend this foundation towards scalable robot manipulation via large-scale empirical studies, novel architectural contributions to temporal reasoning, and a unified modular safety framework with formal guarantees. All research will be implemented in the JAX ecosystem for TPU compatibility and validated on the ManiSkill3 GPU-parallelised robot manipulation benchmark suite. We request $100,000 in unrestricted funding and TPU compute access.

PROBLEM STATEMENT

The state of the art in learning-based robot manipulation faces four compounding challenges that our group is uniquely positioned to address.

Sample efficiency and the exploitation gap. Offline-to-online RL is a promising paradigm: pre-train on safe demonstrations, then fine-tune online. However, methods that constrain online exploration to a learned CVAE latent manifold — the most principled approach to safe exploration — impose a structural performance ceiling: because the decoder maps the latent space back to actions, the policy can only ever recover actions that the CVAE can reconstruct, and the resulting

exploitation gap is bounded below by the reconstruction error. Our group has characterised this gap formally and proposed a curriculum solution (SPAARS), but extending it to high-dimensional manipulation tasks with complex subtask structure remains an open challenge that requires both algorithmic innovation and large-scale compute.

Dense reward engineering. Even in tasks where dense rewards are available, their mathematical form — curvature, support, monotonicity, how stages are gated and combined — has a decisive effect on whether policy gradient training converges or collapses. This dependency is not captured by potential-based shaping theory, and no systematic empirical study exists. Beyond hand-engineered rewards, vision-language models (VLMs) and large language models (LLMs) present a compelling avenue for automated reward generation, but current approaches lack a principled understanding of when and how such signals can reliably guide RL.

Temporal reasoning in neural architectures. A fundamental and largely unacknowledged limitation of current AI systems is that they possess no intrinsic model of time. An agent can represent sequences, yet it cannot reason about duration, anticipate when a condition will hold, or schedule actions based on temporal context. For long-horizon manipulation, this is not merely a performance limitation: it is a structural gap that prevents agents from generalising across task variants with different temporal structures, and whose implications for the foundations of AI (e.g., claims about machine understanding and generality) have received insufficient critical attention.

Safety and hard-to-model constraints. Deployment of learning-based robots is gated on the ability to enforce constraints that are difficult to specify formally. Such constraints take at least four distinct forms in the robotics literature: cost functions (learnable via inverse constraint learning), logical rules (expressible in signal temporal logic), forbidden state regions (encodable as control barrier functions), and ill-defined human preferences (currently addressed only empirically, without guarantees). No existing framework unifies these paradigms into a single modular safety architecture.

RESEARCH OBJECTIVES

- **[WP1: *Structured Latent Representations for Complex and Long-Horizon RL.*]** Building directly on the SPAARS offline-to-online framework, we aim to extend structured CVAE-based latent representations from short-horizon locomotion tasks to the richer setting of multi-stage robot manipulation. Key research questions include: how to discover and represent long-horizon subtask structure within a unified latent space; how to use keyframe-based subgoal discovery (from video demonstrations) to define intermediate reward signals that make long-horizon sparse-reward tasks tractable; and how the advantage-gated mode selection mechanism of SPAARS generalises when the task structure itself must be inferred rather than given. We will also investigate how a single latent representation can be shared and specialised across a family of related tasks, and whether inverse dynamics models learned jointly with the CVAE — constrained by cycle-consistency — can improve both exploration safety and sample efficiency. The objective is a principled, scalable offline-to-online RL framework for manipulation that retains the formal exploitation-gap guarantees of SPAARS while handling the temporal and hierarchical complexity of real-world tasks.

- **[WP2: *Principled Dense Reward Engineering and Automated Reward Generation.*]** This work package addresses reward design from two complementary directions. From the *engineering* side, we will conduct a systematic, large-scale ablation of dense reward function geometry across manipulation tasks, varying curvature, support, monotonicity, gating mechanisms, and terminal bonus structure, with the objective of identifying the causal mechanisms that govern policy gradient dynamics and distilling the findings into an actionable reward design playbook. We will additionally develop distributional approaches to reward specification, where a human provides an inductive prior on the shape of the reward distribution rather than a point-wise formula, encoded and sampled via a CVAE. From the *generation* side, we will investigate the conditions under which VLM- and LLM-generated reward signals can reliably guide RL training: what prompting strategies, reward normalisation procedures, and feedback mechanisms are necessary

for automated reward generation to be both accurate and stable, and how active learning can be used to minimise the number of model queries required. The overarching objective is a principled, reusable methodology for dense reward design that covers both human-engineered and automatically generated reward signals.

- **[WP3: *Implicit Temporal Reasoning in Neural Architectures.*]** We propose a dedicated research programme on temporal intelligence in learning systems. The central observation is that current deep RL agents — including state-of-the-art transformers and recurrent networks — represent time only as a positional index in a sequence. They cannot reason about duration (how long has this state held?), anticipate temporal transitions (when will this condition become true?), or build plans that respect temporal constraints. This limitation is fundamental: it implies that agents cannot generalise to task variants with different temporal structure, cannot compose skills with temporally varying durations, and cannot satisfy constraints expressed in temporal logic without explicit supervision.

  We will pursue this in three directions. First, we will develop neural architectural primitives for *implicit temporal reasoning*: mechanisms that allow a network to represent and propagate beliefs about time as a structured latent dimension, not merely a counter. Second, we will investigate how such representations improve skill transition in hierarchical agents (directly relevant to WP1) by enabling the agent to estimate when a skill is nearing completion rather than relying on fixed-horizon commitments. Third, we will develop a benchmark suite for evaluating temporal reasoning in robotic agents, and accompany it with a position paper arguing that the current absence of intrinsic temporal models is a critical and underappreciated limitation of machine intelligence — with direct implications for claims about generality and understanding in AI systems.

- **[WP4: *Unified Safety and Constraint Satisfaction with Formal Guarantees.*]** Real-world robot manipulation is governed by constraints that can rarely be expressed in a single formal language. We identify four paradigms that must ultimately be reconciled: (i) *cost functions*, where constraints are learned from demonstrations via inverse constraint learning (analogous to IRL but recovering costs rather than rewards); (ii) *logical rules*, where constraints take the form of signal temporal logic (STL) formulae mined from expert trajectories; (iii) *forbidden sets* from control theory, encoded as control barrier functions (CBFs) that provide hard safety guarantees; and (iv) *human preferences*, which are ill-defined, context-dependent, and difficult to formalise yet routinely enforced by human operators. Our research plan follows two parallel threads. The first uses mechanistic interpretability to probe whether VLMs and vision-language action models (VLAs) already encode safety-relevant information in their representations — and, where they do, to leverage these representations as zero-shot safety labellers for building constraint models from minimal demonstrations. The second develops a modular safety architecture that can accept constraint specifications in any of the four paradigms above and compose them into a unified safety filter with formal guarantees. The overarching research question is: can we build a safety filter or safety module for policies learned from minimal human demonstrations, in a way that provides formal guarantees regardless of which constraint paradigm is most appropriate for the task at hand?

WORKFLOW AND EXPECTED OUTCOMES

All experimental work will use the ManiSkill3 simulation platform, which supports massively parallel GPU/TPU environment evaluation across a wide range of manipulation tasks. Policy learning and neural architecture experiments will be implemented in the JAX ecosystem (JAX, Flax, Optax) for native TPU compatibility; scalable on-policy RL will use the Tunix library. All code, trained models, datasets, and analysis notebooks will be released publicly under open-source licences.

Expected primary outcomes:

1. A generalisation of SPAARS to complex manipulation with formal exploitation-gap guarantees,

evaluated on long-horizon ManiSkill3 tasks and released as an open library.

2. An empirically validated reward engineering playbook and suite of automated reward generation tools for RL practitioners.
3. A set of neural architectural primitives for temporal reasoning, a benchmark suite for temporal intelligence in robots, and an accompanying position paper.
4. A modular, formally guaranteed safety architecture unifying inverse constraint learning, STL mining, latent CBFs, and VLM safety labelling, with open-source implementations and validated on complex manipulation scenarios.
5. Publications at top venues in robot learning, reinforcement learning, and AI safety.

RELATION TO ONGOING WORK ACTIVITIES

Our group's primary ongoing project in this area is **SPAARS** — Safer RL Policy Alignment through Abstract Exploration and Refined Exploitation of Action Space (under review at IROS 2026). SPAARS addresses the offline-to-online RL problem by recognising that methods which constrain online exploration to a learned CVAE latent manifold — while provably reducing gradient variance and keeping exploration within the safe behavioural support of the offline data — impose a structural performance ceiling (the exploitation gap) bounded by the CVAE's reconstruction error. SPAARS resolves this via a curriculum: an initial latent exploration phase that exploits variance reduction and safe behavioural grounding, followed by a transition to raw action space that eliminates the reconstruction bottleneck. The transition is governed by a per-state advantage gate inspired by the Option-Critic termination gradient theorem, where the shared critic selects between latent and raw modes based on state-dependent advantage estimates rather than a global schedule. SPAARS has two instantiations: a standalone CVAE-based variant requiring only unordered (s, a) pairs, and a SPAARS-SUPE variant that pairs SPAARS with OPAL temporal skill pretraining for richer exploration structure on tasks where trajectory chunks are available. It has been validated on D4RL locomotion (hopper, walker) and kitchen-mixed benchmarks.

WP1 directly extends SPAARS to long-horizon manipulation tasks. WP2, WP3, and WP4 are independent research directions that address complementary open problems in the same broad setting of learning-based robot manipulation.

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DATA POLICY

All data generated in this project — training logs, reward curves, policy checkpoints, demonstration datasets, and safety constraint models — will be made publicly available on GitHub and a persistent data repository under an open-source licence within 30 days of corresponding publications. The ManiSkill3 simulator is open-source (Apache 2.0). No personal, medical, or sensitive data is used; all environments are fully synthetic.